

KEMPTVILLE, ON, Canada

WMO: 713000

Lat: 45.005N

Lon: 75.637W

Elev: 326

StdP: 14.52

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-7.4	-20.6	1.8	-11.5	-15.6	2.4	-6.5	19.0	22.0	16.7	17.4	4.8	310	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	87.2	71.9	84.4	70.0	81.6	68.7	75.4	82.5	73.4	80.5	71.4	78.2	8.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	125.5	78.8	71.0	115.8	77.5	69.0	107.9	75.5	39.1	82.8	37.2	80.5	35.4	78.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.0	12.2	DB	-19.9	91.0	5.6	2.6	-23.9	92.9	-27.1	94.4	-30.3	95.8	-34.3	97.7	(4)
(5)			WB	-20.2	78.1	5.5	2.3	-24.1	79.8	-27.3	81.1	-30.4	82.4	-34.3	84.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	15.6	18.5	29.0	43.1	55.5	64.5	69.5	67.8	60.5	47.8	35.8	23.4	
	DBStd	20.91	13.24	11.25	11.14	9.10	7.91	6.69	5.49	5.64	7.54	8.60	9.39	11.69	
	HDD50	4292	1066	881	655	247	34	0	0	0	5	149	431	824	
	HDD65	7940	1531	1301	1116	660	311	91	18	36	177	535	875	1289	
	CDD50	2244	0	0	4	38	204	434	604	553	320	81	6	0	
	CDD65	417	0	0	0	2	16	75	157	124	41	2	0	0	
	CDH74	3569	0	0	2	28	202	660	1362	954	341	16	0	4	
	CDH80	924	0	0	0	3	43	169	395	238	75	0	0	1	
Wind	WSAvg	5.3	5.9	6.0	6.2	6.3	5.4	4.7	4.5	4.3	4.4	5.1	5.6	5.5	
Precipitation	PrecAvg	35.5	2.8	2.2	2.4	3.0	2.8	3.4	3.4	3.2	3.4	3.2	2.7	3.0	
	PrecMax	45.0	5.0	4.2	4.6	6.0	4.9	6.8	7.2	6.9	6.1	6.9	4.2	5.0	
	PrecMin	29.6	1.2	0.8	0.9	0.5	0.9	0.5	0.6	1.4	1.5	0.7	0.7	0.9	
	PrecStd	3.6	1.1	0.9	1.1	1.4	1.1	1.4	1.5	1.0	1.3	1.3	0.9	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	49.0	64.4	77.9	84.7	88.9	91.3	90.0	87.0	76.2	64.0	55.8	
		MCWB	49.1	44.1	54.9	63.5	67.9	73.5	73.7	72.8	72.5	66.0	55.5	51.6	
	2%	DB	44.1	42.3	55.0	70.8	79.7	85.0	87.4	85.5	81.7	70.7	58.9	47.1	
		MCWB	41.9	38.6	47.7	55.6	63.1	71.0	72.2	71.1	69.4	61.6	53.8	44.8	
	5%	DB	38.2	38.1	48.9	64.2	74.9	81.0	84.6	82.3	77.8	66.0	54.3	42.3	
		MCWB	35.7	35.0	42.8	52.8	61.0	68.3	70.2	69.0	67.3	59.0	49.5	40.0	
	10%	DB	34.1	34.8	43.9	58.8	70.4	77.6	81.5	79.5	74.1	61.2	49.8	37.9	
		MCWB	32.3	32.4	38.8	49.4	59.1	66.0	69.0	68.4	65.8	55.4	45.9	35.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	44.5	56.4	64.3	70.7	77.8	79.4	76.9	74.3	68.9	58.2	52.3
			MCDB	50.9	46.4	63.6	73.3	81.8	84.7	86.9	83.7	82.3	73.8	60.4	55.9
2%		WB	41.7	38.9	48.4	59.2	67.1	73.9	75.8	74.4	71.7	63.4	54.1	44.9	
		MCDB	43.6	41.9	52.6	67.0	75.3	81.1	83.2	81.3	78.9	68.7	58.6	47.0	
5%		WB	36.1	35.3	43.7	54.6	64.0	71.1	73.5	72.4	69.3	59.8	50.1	39.9	
		MCDB	37.9	37.5	48.2	61.7	71.2	78.2	80.3	78.9	75.7	64.9	53.5	42.1	
10%		WB	32.4	32.7	39.3	50.6	61.0	68.4	71.2	70.4	67.0	56.3	46.0	35.8	
		MCDB	34.1	34.7	43.1	57.6	67.1	74.5	78.0	76.4	72.7	60.5	49.3	37.5	
Mean Daily Temperature Range	5% DB	MDBR	16.7	17.7	17.4	19.4	20.7	19.7	20.2	19.9	20.6	16.7	15.2	13.9	
		MCDBR	18.5	18.7	22.6	29.2	27.5	24.0	23.4	23.0	23.7	22.7	21.5	17.6	
		MCWBR	17.4	16.3	15.8	17.4	14.6	12.4	11.2	11.2	13.0	14.6	16.2	16.0	
	5% WB	MCDBR	17.9	17.8	21.1	24.3	22.7	20.3	19.6	19.2	21.2	20.4	20.5	17.4	
		MCWBR	17.2	16.2	16.3	17.0	14.3	12.1	11.7	11.4	12.8	14.8	16.6	16.4	
Clear-Sky Solar Irradiance	taub	0.274	0.291	0.321	0.347	0.380	0.414	0.422	0.406	0.367	0.341	0.308	0.280		
	taud	2.395	2.350	2.400	2.406	2.352	2.267	2.248	2.297	2.399	2.469	2.499	2.430		
	Ebn at Noon	269	285	290	290	282	272	267	268	270	263	255	254		
	Edh at Noon	23	29	32	35	38	42	42	39	32	26	21	20		
All-Sky Solar Radiation	RadAvg	436	734	1151	1409	1656	1789	1826	1609	1268	749	464	325		
	RadStd	48	72	96	157	167	141	154	75	107	69	52	33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page